Problem Set 2

Industrial Organization

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2026-10-23

Part 0: Empirical Target

Object of Interest

Our main empirical objective is to **quantify the welfare gains associated with the availability of gluten-free noodle options on college campuses**. Specifically, we aim to estimate the change in student welfare relative to a counterfactual market in which none of the chains offers gluten-free options. Our object of interest is the change in consumer surplus resulting from the introduction of gluten-free options, both on average and separately across groups defined by students' observable characteristics. This allows us to assess not only the overall benefits of expanding the available choice set, but also how these benefits are distributed across different types of students.

Demand Estimation and Consumer Preferences

To quantify these welfare gains, we will estimate a discrete-choice demand model based on a random utility framework. Student utility will depend on product prices and observable characteristics, such as spiciness, wheat content, the vegetable-to-meat ratio, and calories, as well as unobserved product and consumer characteristics. We begin with the logit specifications in Part II and then allow for heterogeneous preferences through random-coefficient models in Part III.

Price Variation and Product Substitution

A key challenge is that observed prices may be endogenous if stores adjust prices in response to unobserved product attractiveness. We will therefore investigate whether local input-cost shocks provide suitable instruments for product prices. Specifically, variation in local prices of wheat, vegetables, and meat, combined with differences in product composition, may generate exogenous variation in production costs and help identify price sensitivity. We will also characterize substitution patterns across products. Observed market shares will inform the estimation of demand, while the hypothetical second-choice responses in the survey can provide additional information on substitution between alternatives.

Counterfactual and Welfare Measurement

Using the estimated demand model, we will simulate a counterfactual in which all gluten-free products are removed from the choice set, while the prices and characteristics of the remaining products are held constant. Holding the prices of the remaining products fixed is a simplifying assumption that isolates the direct welfare effect of expanding the available choice set and facilitates computation. It abstracts from potential equilibrium price adjustments following the introduction or removal of gluten-free products.

We will measure welfare changes using the model-implied change in consumer surplus, accounting for the estimated price sensitivity and, where applicable, heterogeneity in preferences. Finally, we will examine heterogeneity in welfare gains across student groups, particularly athletes and graduate students. Comparing results across the logit and random-coefficient specifications will allow us to assess how assumptions about preference heterogeneity affect the estimated benefits of gluten-free options.

Part 1: Empirical Evidence, Early Exhibits

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Figure 1 Panel A documents the evolution of noodle market shares between 2001 and 2020. Traditional noodles account for all of the observed noodles in the market before the introduction of gluten-free (GF) options in 2011. Following their introduction, GF noodles gain market share, with a further increase around 2016, when additional chains begin offering GF alternatives. Over the same period, the market share of traditional noodles declines. These patterns are consistent with a change in the composition of noodle demand as GF alternatives become available, although the figure alone cannot establish that their introduction caused the decline in traditional noodle shares. Panel B reproduces the same analysis by noodle chain. Since YourNoodles was the first to introduce GF options we observe its market share rising in 2011, but after all brands introduce it in 2016 it actually declines and others go up, suggesting BestNoodles and AllNoodles might have more attracting options at this point.

Figure 2 examines whether GF choices vary across student groups over time. The share of GF choices increases for both athletes and non-athletes, as well as for graduate students and non-graduates since the first introduction of a GF option in 2011, particularly after 2015. Non-athletes seem to consistently consume more GF options, while for Graduate vs non Graduate-students this relationship varies depending on the year. The overall patterns suggest potential heterogeneity in adoption patterns. However, these comparisons are descriptive and may also reflect changes in the composition of the surveyed students and campuses over time.

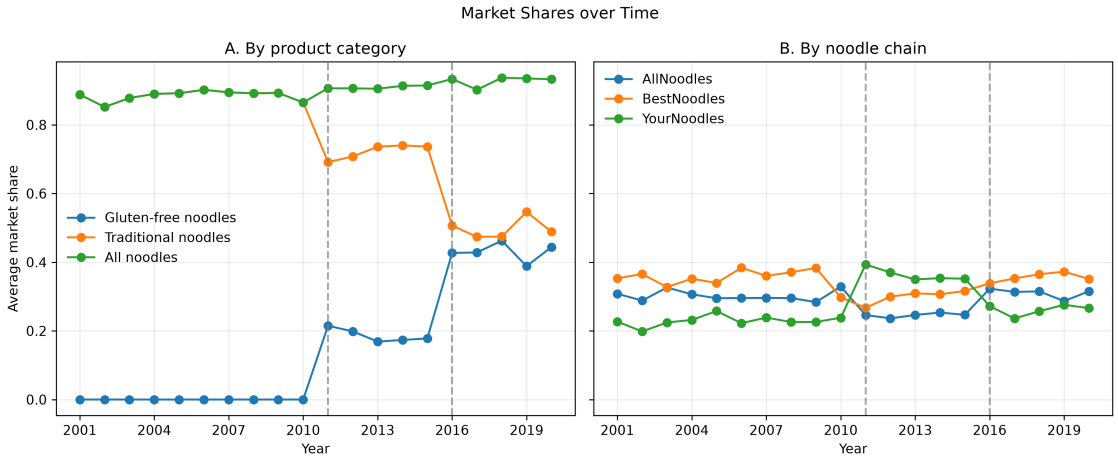


Figure 1: Market shares by product category and noodle chain

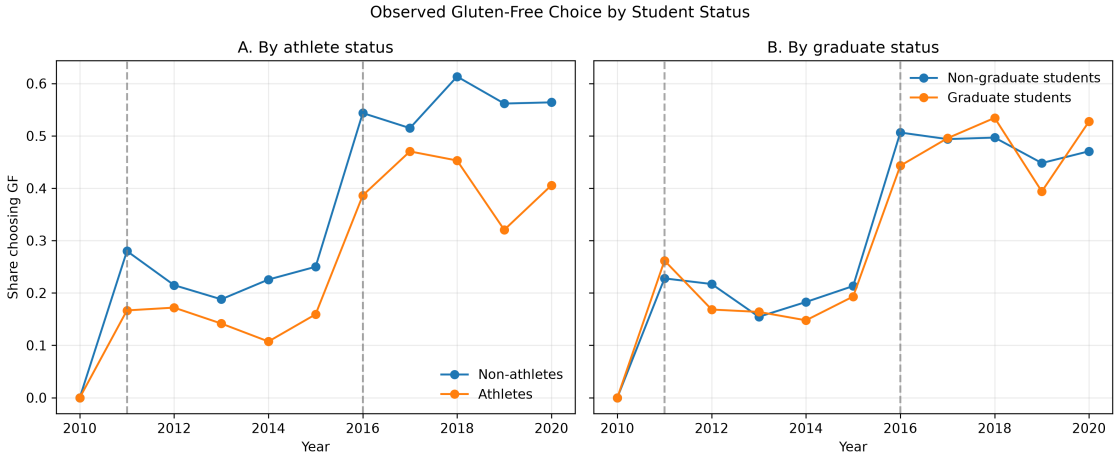


Figure 2: Observed gluten-free choice by athlete and graduate status

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We present two main pricing regression exercises - and two complementary. Our main exercises refer to Regressions 1 and 4 (baseline versus adding cost and market shocks) but we also add intermediate results adding both sets of shocks individually to better assess their effect in the estimation.

Regression 1 — Baseline

$$Price_{jmt} = \beta_0 + X'_{jmt}\beta + u_{jmt}.$$

Regression 2 — Market shocks

$$Price_{jmt} = \beta_0 + X'_{jmt}\beta + \gamma' C_{mt} + u_{jmt}.$$

Regression 3 — Input-cost shocks

$$Price_{jmt} = \beta_0 + X'_{jmt}\beta + \delta' Z_{jmt} + u_{jmt}.$$

Regression 4 — All cost shocks

$$Price_{jmt} = \beta_0 + X'_{jmt}\beta + \gamma' C_{mt} + \delta' Z_{jmt} + u_{jmt}.$$

Here, j identifies the product (brand and dish), m the campus, and t the year. The vectors enter the specifications, and appear in the regression table, in the following order:

- $X_{jmt} = (\text{SpicyIndex}, \text{WheatContent}, \text{Veg2meat}, \text{Calories})'$ contains the observed product characteristics.
- $C_{mt} = (\text{WageIDX}, \text{ReEsIDX})'$ contains market-level cost shifters: the local wage and real-estate indices.
- $Z_{jmt} = (\text{wheat_shock}, \text{vegetable_shock}, \text{meat_shock})'$ contains product-market-specific input-cost shocks, constructed from local input prices and the product's exposure to wheat, vegetables, and meat.

The vectors β, γ, and δ are the corresponding coefficient vectors, and u_{jmt} is the unobserved component of price. Standard errors are heteroskedasticity robust (HC1), so all estimates are interpreted as conditional associations rather than causal effects.

Table 1 reports regression results. Column (1) shows the OLS relationship between menu prices and product characteristics. Since prices may be correlated with unobserved product characteristics, we interpret the coefficients as conditional associations rather than causal effects. Wheat content is positively and statistically significantly associated with price, whereas the coefficients on spiciness, vegetable-to-meat ratio, and calories are not statistically significant. Advancing to columns (2), (3), and (4) regression results suggest that both market-level cost shifters and product-market-specific input-cost shocks contribute to explaining price variation - separately and together. In all specifications shocks are jointly statistically significant at the 1% level in F-stat tests. Comparing the F-stats within specifications results suggest that product-market-specific input-cost shocks account for a larger share of the variation explained by the pricing regressions than market-level cost shifters alone. However, these results establish explanatory relevance in the pricing regressions, not instrument strength in the demand estimation or instrument exogeneity. The latter requires further assessment in the instrumental-variables estimation and an economic justification for the exclusion restrictions.

Table 1: Pricing Regressions with Market and Input-Cost Shocks

Variable	(1) Baseline	(2) Market shocks	(3) Input-cost shocks	(4) All cost shocks
const	0.6757***	0.2798*	0.7465***	0.3776***
	(0.1641)	(0.1646)	(0.1363)	(0.1336)
SpicyIndex	-0.0419	-0.0296	-0.0196	-0.0082
	(0.0557)	(0.0537)	(0.0464)	(0.0442)
WheatContent	0.6555***	0.6731***	-0.2233***	-0.1955***
	(0.0619)	(0.0582)	(0.0624)	(0.0576)
Veg2meat	0.0667	0.0555	-0.2678**	-0.2638***
	(0.1065)	(0.1021)	(0.1051)	(0.1001)
Calories	-0.0166	0.0218	-0.0999	-0.0598
	(0.1809)	(0.1747)	(0.1473)	(0.1410)
WageIDX	—	0.1750***	—	0.1711***
		(0.0177)		(0.0175)
ReEsIDX	—	0.0515***	—	0.0365**
		(0.0182)		(0.0143)
wheat_shock	—	—	0.6989***	0.6899***
			(0.0310)	(0.0288)
vegetable_shock	—	—	0.2175***	0.2051***
			(0.0407)	(0.0395)
meat_shock	—	—	-0.0004	-0.0005
			(0.0015)	(0.0015)
Observations	2690	2690	2690	2690
R-squared	0.0479	0.1490	0.2918	0.3861
F-statistic (included shocks)	—	50.45	176.03	145.48
F-test p-value	—	<0.001	<0.001	<0.001

Notes: Dependent variable: price. Heteroskedasticity-robust standard errors are in parentheses. The F-test tests whether the shocks included in each specification are jointly equal to zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reduced-Form Estimations

For our reduced form exhibits we begin wit Figure 3 which reports the event-study estimates around the introduction of gluten-free options. The following figures then describe market shares overall and heterogeneity in observed gluten-free choices.

To obtain quasi-experimental variation in gluten-free (GF) availability, we exploit the staggered adoption across chains: YourNoodles introduced GF in 2011 in all campuses, while AllNoodles and BestNoodles only did so in 2016. A two-way event study using all years would lack a clean control group after 2016, when every chain is treated. We therefore restrict the sample to 2001–2015, where AllNoodles and BestNoodles serve as not-yet-treated controls. We regress chain-campus-year market share on event-time indicators for YourNoodles (reference period k = −1), with chain fixed effects and campus-by-year fixed effects, clustering standard errors by campus. The campus-year fixed effects absorb any shock common to all chains in a market.

The pre-adoption coefficients are small and statistically indistinguishable from zero, supporting parallel trends. After 2011, YourNoodles' share rises by about 0.21 on impact and settles at roughly 0.15 over the following years, with all post-period coefficients significant. This is a relative effect: it measures YourNoodles' share against the other chains in the same campus-year, not the aggregate effect of GF on total noodle consumption.

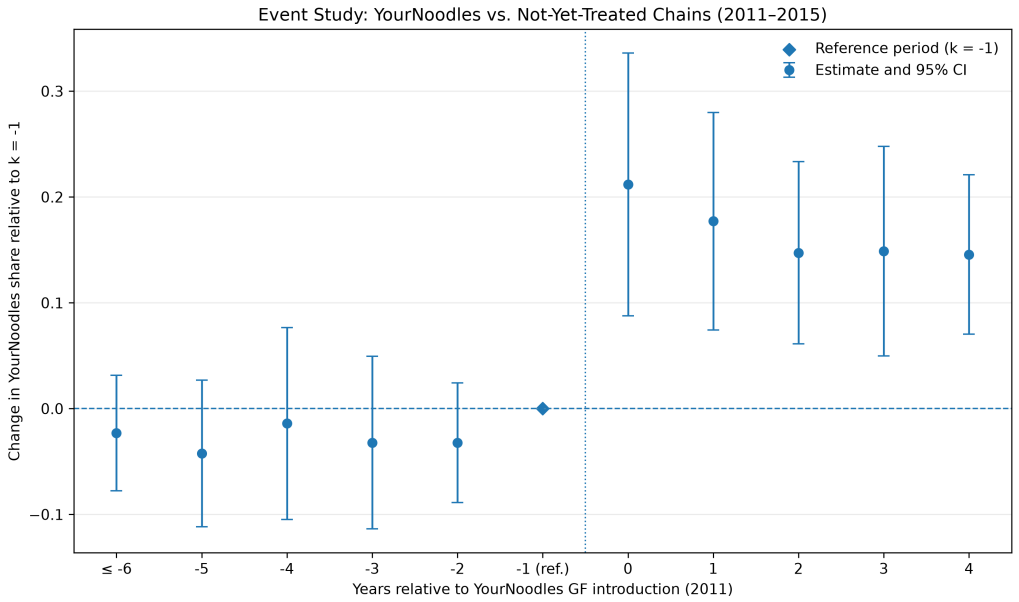


Figure 3: Event study: noodle market share around gluten-free introduction

On another exhibit, Table 2 documents substitution patterns across chains and product categories. Among respondents whose observed choice was gluten-free, the share reporting a traditional noodle dish as their hypothetical second choice ranges from 54.7% for AllNoodles to 80.8% for YourNoodles. Conversely, among respondents whose observed choice was traditional, 24.5%–31.8% report a gluten-free dish as their second choice, depending on the chain. These patterns suggest that gluten-free and traditional noodles can be substitutes, but the extent of substitution varies across chains. They also indicate that product category alone does not fully characterize consumers' second-choice patterns. Since these are stated hypothetical choices, they should not be interpreted as actual switching behavior or as direct estimates of substitution following the removal of all gluten-free products.

Table 2: Second-Choice Summary by Chain

Observed choice	Second choice: GF	Second choice: Traditional	N
AllNoodles – GF	45.3%	54.7%	752
AllNoodles – Traditional	31.8%	68.2%	2,382
BestNoodles – GF	39.4%	60.6%	836
BestNoodles – Traditional	27.3%	72.7%	2,750
YourNoodles – GF	19.2%	80.8%	1,527
YourNoodles – Traditional	24.5%	75.5%	1,784

Part 2

5.

Dividing everything by σ,

$$\frac{1}{\sigma} u_{ijmt} = -\frac{1}{\sigma} p_{jmt} + \frac{\mu_j}{\sigma} + \frac{\beta'}{\sigma} X_{jmt} + \frac{\xi_{jmt}}{\sigma} + \varepsilon_{ijmt}.$$

If we define $\alpha = 1/\sigma$ and rescale the coefficients and utility, we get

$$\tilde{u}_{ijmt} = -\alpha\, p_{jmt} + \tilde{\mu}_j + \tilde{\beta}' X_{jmt} + \tilde{\xi}_{jmt} + \varepsilon_{ijmt}.$$

From now on, this is the form we use, dropping the tildes to have everything rescaled.

6.

First, let $\delta_{jmt} = -\alpha\, p_{jmt} + \mu_j + \beta' X_{jmt} + \xi_{jmt}$. Choice probabilities are

$$s_{jmt} = P\Big\{\delta_{jmt} + \varepsilon_{ijmt} \geq \delta_{\ell mt} + \varepsilon_{i\ell mt}\ \forall \ell\Big\} = \frac{e^{\delta_{jmt}}}{\sum_\ell e^{\delta_{\ell mt}}},$$

by the properties of the Gumbel distribution. In the data, we see that the shares add up to less than one. We consider that those who are not consuming noodles are picking the outside option of good 0, which has utility ε_{i0mt}, and mean utility $\delta_{0mt} = 0$. By the same logit formula, the outside option has market share $1/(\sum_\ell e^{\delta_{\ell mt}})$. Dividing one by the other,

$$\frac{s_{jmt}}{s_{0mt}} = e^{\delta_{jmt}} \implies \log(s_{jmt}) - \log(s_{0mt}) = \delta_{jmt}.$$

Using the definition of δ, we get the linear regression

$$\log(s_{jmt}) - \log(s_{0mt}) = -\alpha\, p_{jmt} + \mu_j + \beta' X_{jmt} + \xi_{jmt}.$$

Under the assumption that $\mathbb{E}\Big[\xi_{jmt} \mid p_{jmt}, X_{jmt}\Big]$, the parameters can be unbiasedly recovered using OLS. As each good has a different μ_j, we include good fixed effects. Estimates are in the first column of the table below.

Table 3: Demand Estimates

	OLS	IV	
Model:	(1)	(2)	(3)
Variables			
Price	-0.638***	-0.826***	-0.829***
	(0.006)	(0.013)	(0.011)
Spicy	-0.063	-0.047	-0.047
	(0.079)	(0.093)	(0.093)
Wheat	0.144***	0.268***	0.270***
	(0.051)	(0.061)	(0.061)
Veg2Meat	-0.029	-0.020	-0.019
	(0.036)	(0.042)	(0.043)
Calories	0.134*	0.155*	0.156*
	(0.072)	(0.084)	(0.084)
Cost-shifter IVs	No	Yes	Yes
Interaction IVs	No	No	Yes
Fixed-effects			
dish	Yes	Yes	Yes
Fit statistics			
Observations	2,800	2,800	2,800
R^2	0.82842	0.45422	0.56939
Within R^2	0.81054	0.39738	0.52454

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.

Now we assume that the characteristics are exogenous, but that prices can be exogenous. We had to do some data cleaning to get all the university names to match in both datasets. We use the variables in Z as instruments for price. To obtain more variation, we also add interactions between Z and how important it is for a good (for example, Wheat Content $\times$ Wheat Costs). This is so that we also have variation across universities, and not only over time. Results are in the same table as before.

As we would expect prices to be positively correlated to positive demand shocks, the price coefficient in OLS should have an upward bias, being closer to zero. This is exactly what we see: the price coefficient becomes more negative with instruments.

8.

Consumer surplus for a given scenario, at a given university and time, takes the form

$$\text{CS} = \frac{1}{\alpha} \log \left(\sum_{\ell} e^{\delta_{\ell m t}} \right).$$

These can be easily computed by plugging in the $\hat{\delta}$'s, which are the predicted values from the regressions. We simulate a counterfactual where gluten-free options were never introduced. We assume that, in that world, prices and everything else would have remained constant. We create a function that takes in the data and a vector or parameters, and outputs the welfare of each university and time. For the counterfactual, we simply drop gluten-free alternatives.

We see, in the figure below, that, while the counterfactual consumer surplus remains flat, the actual estimated one has two increases. The first is in 2011, when gluten-free options started to be introduced, and the second in 2016, when more options became available. The same pattern holds for OLS and for IV, but the scale is different.

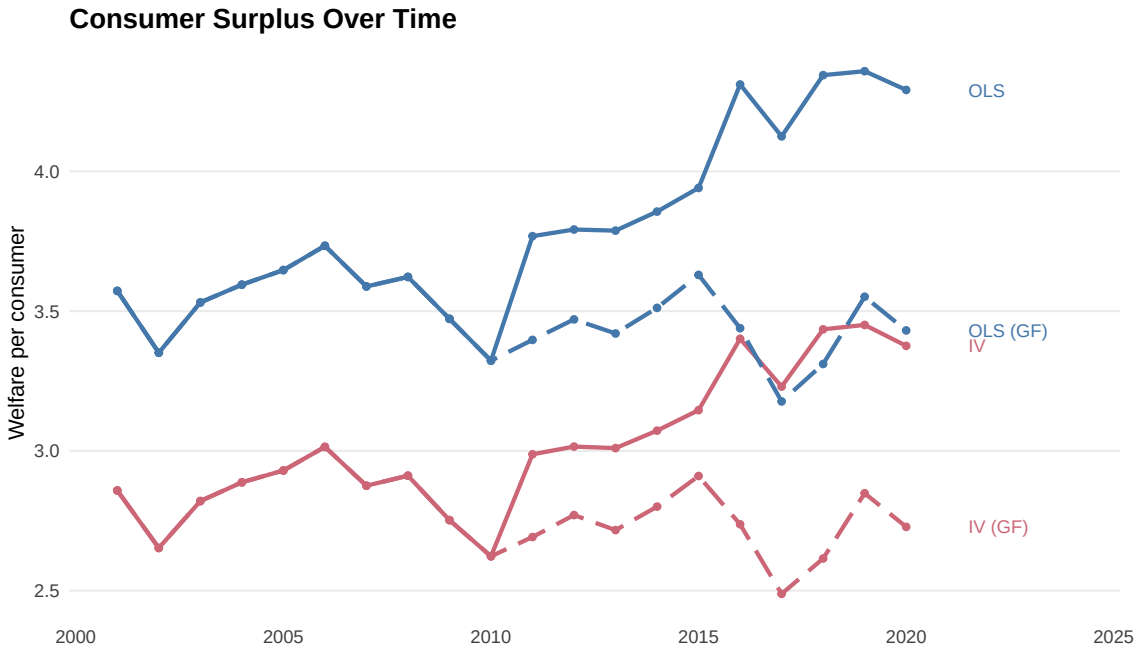


Figure 4: Consumer welfare over time

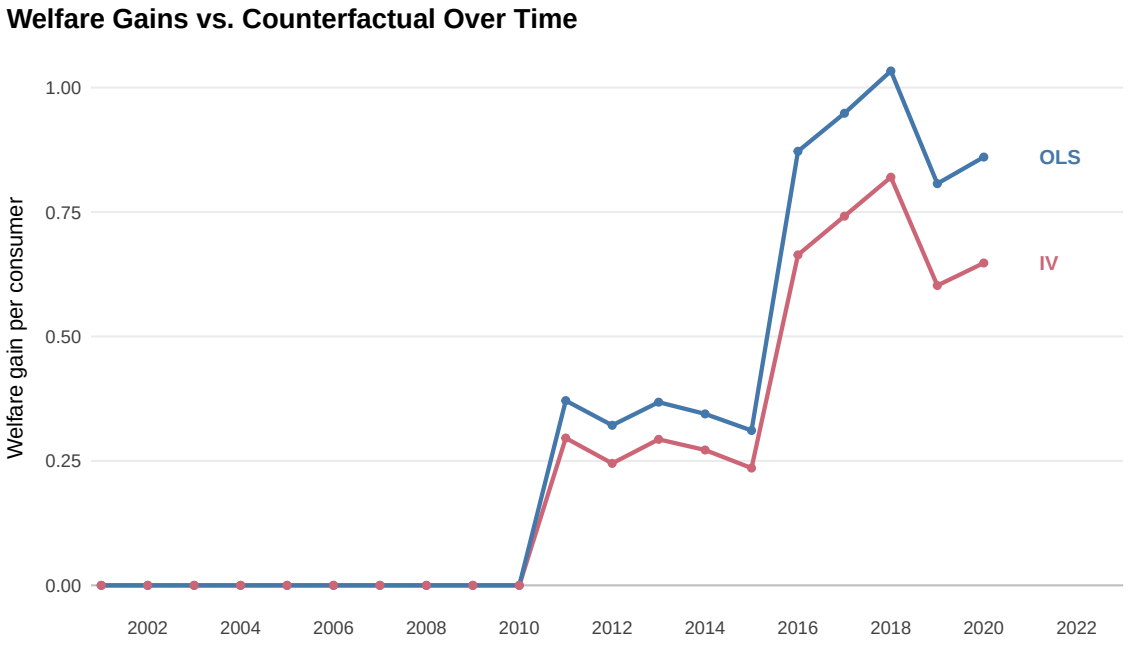


Figure 5: Welfare gains over time

Regressing the welfare gains on the characteristics of each university per year, we cannot find any heterogeneity depending on the type of student population.

Table 4: Welfare Gains and Student Composition

	2011–2015	2016 onward
Panel A: OLS		
Constant	0.341***	0.731***
	(0.065)	(0.122)
Share of Athletes	-0.008	-0.013
	(0.073)	(0.123)
Share of Graduates	0.013	0.364*
	(0.113)	(0.198)
Panel B: IV		
Constant	0.263***	0.522***
	(0.063)	(0.116)
Share of Athletes	-0.015	0.003
	(0.070)	(0.117)
Share of Graduates	0.026	0.348*
	(0.109)	(0.189)

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9.

The elasticities implied by logit are

$$\eta_{jk} = p_k \frac{\partial \log s_j}{\partial p_k} = p_k \frac{\partial}{\partial p_k} \left[\delta_j - \log \sum_{\ell} e^{\delta_\ell} \right].$$

So, elasticities are

$$\eta_{jk} = p_k \Big[-\alpha \mathbb{I}\{j = k\} + s_k \Big].$$

They depend only on data and on our estimate $\hat{\alpha}$. We compute the elasticity for each university and time, then average them out, using both the OLS and IV estimates. Below are our estimates of average elasticities, for the three time periods.

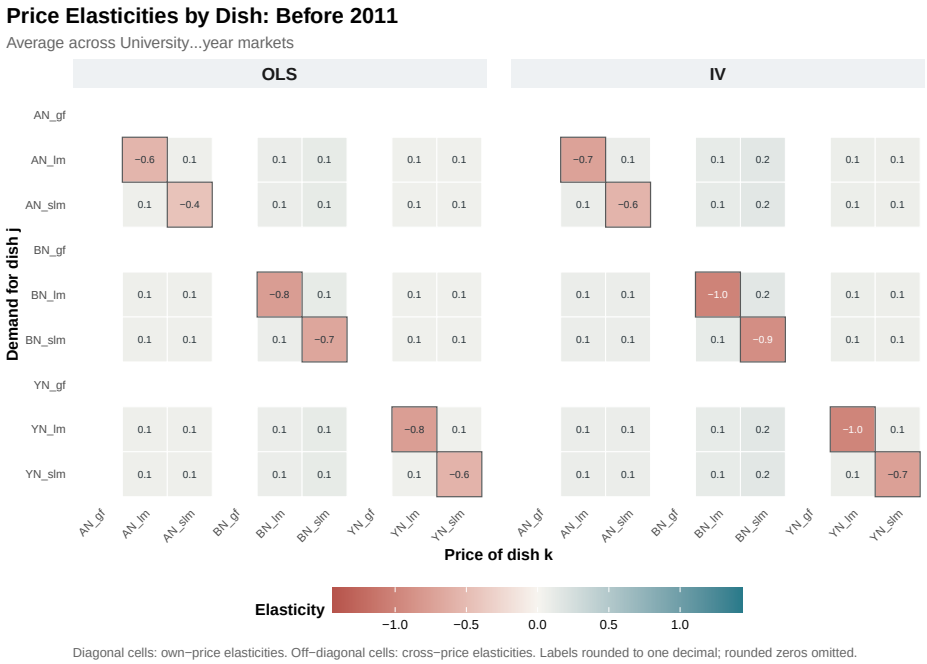


Figure 6: Price elasticities, before 2011

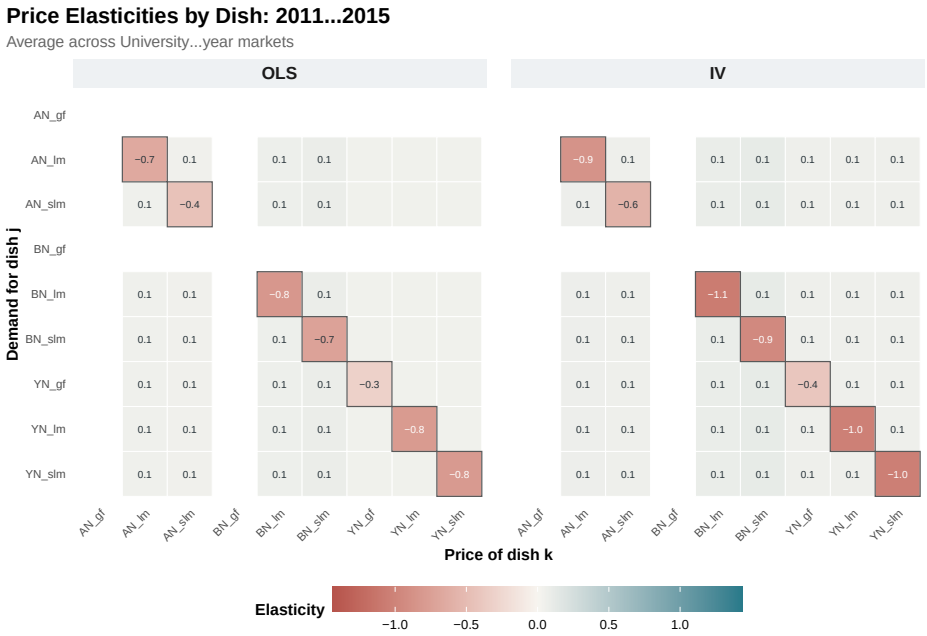


Figure 7: Price elasticities, 2011-2015

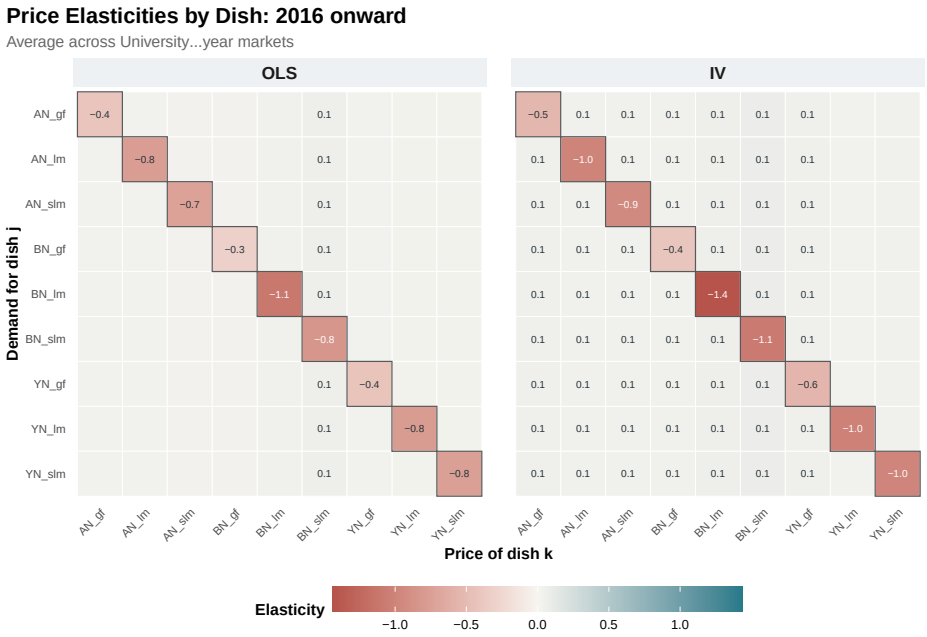


Figure 8: Price elasticities, 2016 onward

Because the market is relatively evenly split and the logit elasticities depend only on market shares and not characteristics, most of the cross-price elasticities sit at around 0.1. This is not intuitive, as the gluten-free options should be closer substitutes among each other, for example. In terms of policy, this implies that the one company producting gluten-free noodles between 2011 and

2015 did not actually hold much market power, as consumers would be willing to easily substitute between gluten-free and gluten-containing dishes. Given this demand, a counterfactual where a single company became a monopolist of gluten-free options likely would not have a large spike in prices, as that would push consumers towards the other alternatives.

10.

Allowing for there to be two nests, $g = 0, 1$, we use the utility specification

$$u_{ijmt} = \delta_{jmt} + \zeta_{imt} + (1-\rho)\varepsilon_{ijmt}, \quad \text{for } j \neq 0$$

where $\delta_{jmt} = -\alpha\, p_{jmt} + \mu_j + \beta' X_{jmt} + \xi_{jmt}$, and the outside option, in a different nest, has utility $u_{i0mt} = \varepsilon_{i0mt}$. The resulting expression for market shares is

$$\log(s_j) - \log(s_0) = \delta_{jmt} + \rho\, \log(s_{j|1}), \quad \text{where } s_{j|1} = \frac{s_j}{1 - s_0}.$$

This means we get a similar regression specification, with one extra term. Besides the cost-shifters we had before, we also add a third specification where we add the number of other elements in the next as an instrument.

Table 5: Nested Logit Demand Estimates

	OLS		IV	
Model:	(1)	(2)	(3)	(4)
Variables				
Price	-0.477***	-0.861***	-0.813***	-0.838***
	(0.008)	(0.032)	(0.021)	(0.016)
ρ	0.355***	-0.120	0.039	-0.022
	(0.014)	(0.100)	(0.044)	(0.027)
Spicy index	-0.067	-0.047	-0.047	-0.046
	(0.071)	(0.095)	(0.092)	(0.094)
Wheat content	0.116**	0.265***	0.268***	0.271***
	(0.046)	(0.062)	(0.060)	(0.061)
Vegetable-to-meat	-0.038	-0.018	-0.020	-0.019
	(0.033)	(0.043)	(0.042)	(0.043)
Calories	0.101	0.164*	0.152*	0.158*
	(0.065)	(0.087)	(0.084)	(0.085)
Cost-shifter IVs	No	Yes	Yes	Yes
Interaction IVs	No	No	Yes	Yes
Nest-size IV	No	No	No	Yes
Fixed-effects				
dish	Yes	Yes	Yes	Yes
Fit statistics				
Observations	2,800	2,800	2,800	2,800
R^2	0.86088	0.45435	0.56945	0.56920

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Addings instruments, ρ seems to be very close to 0, which means that the model collapses to a standard logit, and the nest doesn't really help. If we run the welfare exercises again for a ρ very close to zero, the results will be basically the same as in the standard logit case. Instead, we use the naive OLS estimates, since it is the only one with a ρ that is significant and positive. Here, the welfare also depends on ρ:

$$\text{CS} = \frac{1}{\alpha} \log\left(1 + \left(\sum_{\ell \neq 0} e^{\delta_{\ell m t}/(1-\rho)}\right)^{1-\rho}\right).$$

And the elasticities are

$$\eta_{jk} = \alpha p_k \left[s_k + \frac{\rho}{1-\rho} s_{k|1} - \frac{1}{1-\rho} \mathbb{I}\{j = k\}\right].$$

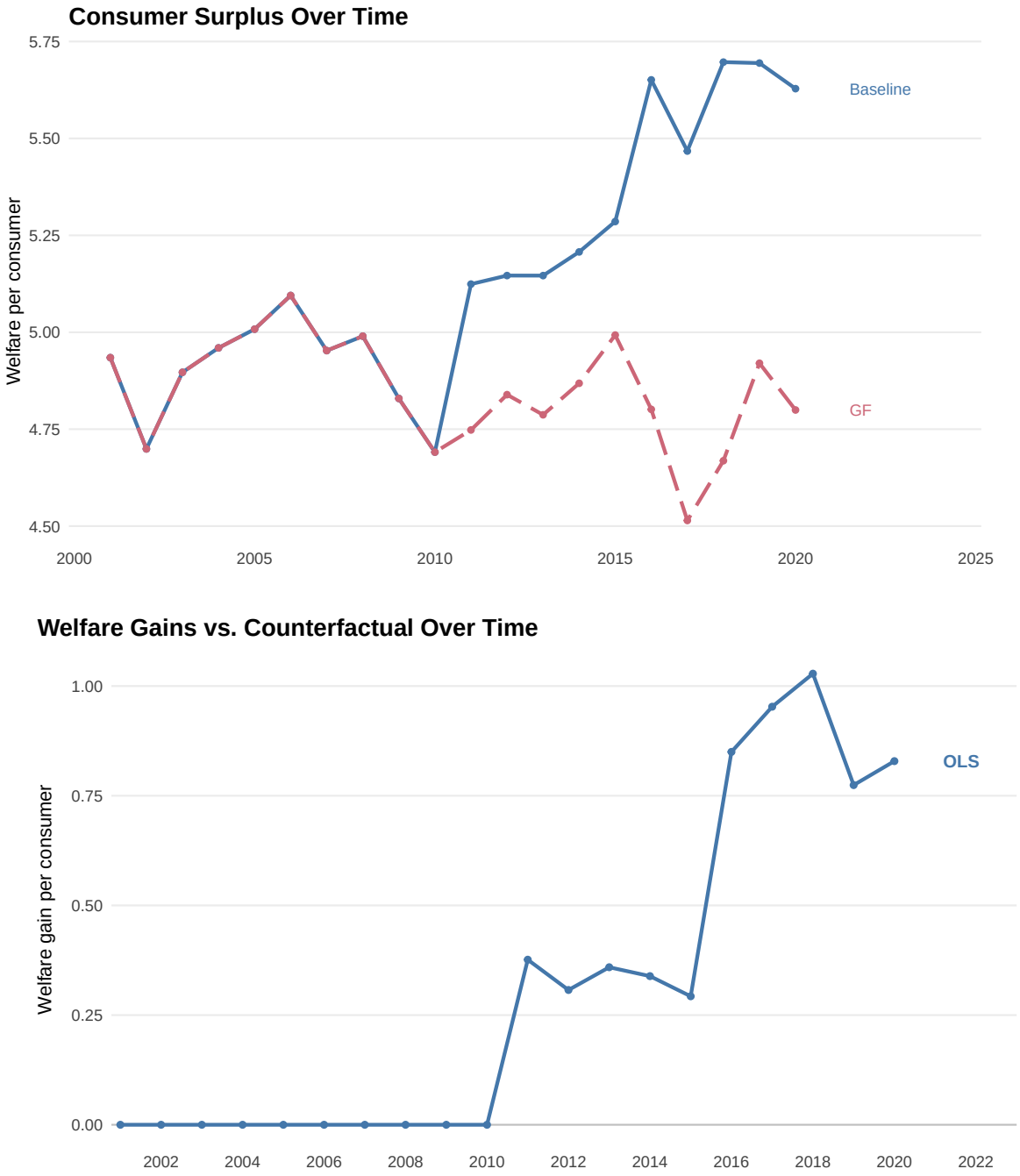


Figure 9: Welfare gains over time

Table 6: Welfare Gains and Student Composition

	2011–2015	2016 onward
Constant	0.331***	0.670***
	(0.078)	(0.138)
Share of Athletes	−0.016	0.020
	(0.088)	(0.139)
Share of Graduates	0.026	0.420*
	(0.136)	(0.225)

Dependent variable: nested-logit welfare gain. OLS demand estimates are used to construct welfare. Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

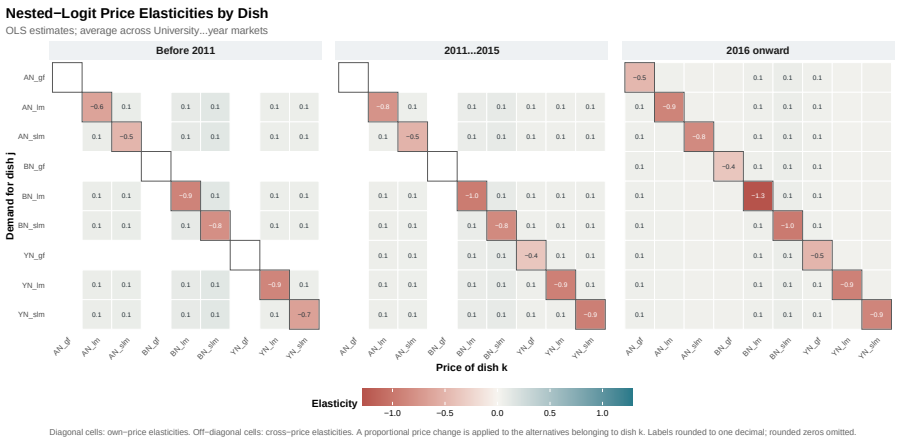


Figure 10: Elasticities

Overall, results are very similar to using the OLS estimator for standard logit.

11.

The Veg2Meat variable fluctuates a lot within dishes, and none of the dishes always have a ratio of 1. Therefore, none of the dishes are always vegetarian. Because of this, we use three nests: one for the gluten-free goods, one for the rest of the noodles, and one for the outside option. For simplicity, we focus on the specification with the most instruments. Here, the regression specification is

$$\log(s_j) - \log(s_0) = \delta_{jmt} + \rho \log(s_{j|g(j)}), \quad \text{where } s_{j|g} = \frac{s_j}{\sum_{\ell \in g(j)} s_\ell},$$

and the outside option just has utility ε_{i0mt}.

Table 7: Nested Logit Demand: Three Nests

Model:	(1)
Variables	
Price	-0.830***
	(0.015)
ρ	-0.003
	(0.041)
Spicy index	-0.047
	(0.093)
Wheat content	0.270***
	(0.061)
Vegetable-to-meat	-0.019
	(0.043)
Calories	0.156*
	(0.085)
Cost-shifter instruments	Yes
Interaction instruments	Yes
Nest-size instrument	Yes
Fixed-effects	
dish	Yes
Fit statistics	
Observations	2,800
R^2	0.56932

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

The ρ we find is, once again, very close to zero, which means the ensts add very little information. The welfare exercises will yield pretty much the same results as we had before. The price coefficient is also very close to the one we had for simple logit.

Part 3

12. Models with Random Coefficients

We estimate the four specifications in Question 12 using the 2,800 product observations from 400 campus year markets. As in Part II.5, normalized utility is

$$u_{ijmt} = \alpha p_{jmt} + \mu_j + x'_{jmt}\beta_i + \xi_{jmt} + \epsilon_{ijmt}, \qquad \alpha < 0,$$

where x_{jmt} contains spicy index, wheat content, vegetable to meat ratio, and calories, and ϵ_{ijmt} is standard type-I extreme value. The outside option has zero normalized utility. We estimate

$$
\begin{aligned}
&(a) \quad \beta_i \sim N(\bar{\beta}, \Sigma), \quad \Sigma \text{ diagonal},\\
&(b) \quad \beta_i = \beta^A A_i + \beta^G G_i,\\
&(c) \quad \beta_i \sim N(\beta^A A_i + \beta^G G_i, \Sigma), \quad \Sigma \text{ diagonal},\\
&(d) \quad \beta_i \sim N(\beta^A A_i + \beta^G G_i, \Sigma),
\end{aligned}
$$

where model (d) allows correlations among wheat, veg/meat, and calorie tastes but not with spicy tastes. We use the exact mean structure specified in the question, without adding a common mean to models (b)–(d). Since the campus year data contain only the marginal shares of athletes and graduate students, we assume independence between these two statuses within markets.

Following Berry, Levinsohn, and Pakes (1995) and Nevo (2000, 2001), we invert simulated choice shares to recover market mean utilities and estimate linear coefficients by IV within two step GMM. Instruments consist of exogenous characteristics, dish indicators, local cost indices, cost characteristic interactions, and other products' characteristics (31 independent columns). We use 512 fixed simulation draws for normally distributed tastes. The recovered mean utilities match the observed shares to numerical precision.

Table III.1. Price coefficients, mean preferences, taste standard deviations

Parameter	(a)	(b)	(c)	(d)
Price	-0.850	-0.851	-0.882	-0.921
Mean spicy	-0.024	–	–	–
Mean wheat	0.293	–	–	–
Mean veg/meat	0.036	–	–	–
Mean calories	0.161	–	–	–
SD spicy	0.003	–	0.000	0.592
SD wheat	0.272	–	0.295	0.356
SD veg/meat	0.591	–	0.847	0.662
SD calories	0.001	–	0.004	0.208

Table III.2. Athlete and graduate mean taste coefficient

Characteristic	(b) Athlete	(b) Grad	(c) Athlete	(c) Grad	(d) Athlete	(d) Grad
Spicy	0.006	0.039	-0.606	0.566	-0.638	0.548
Wheat	0.234	0.305	0.164	0.400	0.352	0.240
Veg/meat	-1.327	1.539	-0.316	0.507	-0.463	0.576
Calories	1.524	-1.290	1.547	-1.334	1.450	-1.231

Price coefficients are negative in all four models. Athletes generally have higher calorie preferences and lower veg/meat preferences, while the graduate coefficients show the opposite pattern. Model (a) estimates more unobserved dispersion in veg/meat and wheat tastes than in spicy or calorie tastes. For model (d), the estimated covariance matrix, in the characteristic order above, is

$$\widehat{\Sigma}_{(d)} = \begin{pmatrix} 0.350530 & 0 & 0 & 0 \\ 0 & 0.126523 & 0.235538 & 0.074069 \\ 0 & 0.235538 & 0.438482 & 0.137889 \\ 0 & 0.074069 & 0.137889 & 0.043382 \end{pmatrix}.$$

Its wheat veg/meat correlation is approximately one, and the other two correlations are about 0.9998. This covariance is numerically singular, so the individual correlations are not reliably identified even though the optimization converged.

13. Welfare effects of Gluten Free options

We measure the compensating variation from introducing gluten free dishes by comparing the observed choice set $\mathcal{J}_{mt}$ with $\mathcal{J}_{mt} \setminus GF$, keeping the other prices and characteristics fixed. For student type h and random taste draw r,

$$\Delta CS_{rhmt} = \frac{1}{-\widehat{\alpha}} \left[\log \left(1 + \sum_{j \in \mathcal{J}_{mt}} e^{V_{jrhmt}} \right) - \log \left(1 + \sum_{j \in \mathcal{J}_{mt} \setminus GF} e^{V_{jrhmt}} \right) \right].$$

We average across taste draws, demographic weights, and campus years. Welfare is in units of the Price variable per lunch choice; markets receive equal weight because their student counts are unavailable.

Table III.3. Mean welfare gain from gluten free dishes

Period	(a)	(b)	(c)	(d)
2001–2010	0	0	0	0
2011–2015	0.256	0.257	0.249	0.239
2016–2020	0.705	0.708	0.685	0.666
2011–2020	0.480	0.482	0.467	0.453

Table III.4. Welfare gain by student type, 2011–2020

Student type	(a)	(b)	(c)	(d)
Neither	0.480	0.525	0.536	0.523
Athlete only	0.480	0.607	0.480	0.434
Graduate only	0.480	0.389	0.412	0.430
Both	0.480	0.414	0.411	0.395

The welfare gain is zero before gluten free entry, and positive under every model after entry. Gains are larger in 2016–2020, when all three chains offered gluten free dishes, though changes across periods also reflect other market changes. Model (a) implies no demographic differences. In (b)–(d), the group specific gains vary more than aggregate welfare, suggesting that the distribution of benefits depends on how preferences are modeled. These are fixed price counterfactuals, not equilibrium welfare changes. Compared with Part II.8, the random coefficient welfare estimates

are closer to the IV results than the OLS results. The Part II welfare plots imply average gains of approximately 0.34 (OLS) and 0.27 (IV) in 2011–2015, and 0.90 (OLS) and 0.70 (IV) in 2016–2020. The corresponding ranges across our four models are 0.239–0.257 and 0.666–0.708. The nested logit OLS results in Part II.10 are also close to simple-logit OLS, at roughly 0.34 and 0.89. However, the nested logit IV estimates have a nesting parameter close to zero, so they provide little evidence for a different welfare prediction. Part II finds no significant athlete share association with welfare gains, but a weak positive graduate share association after 2016. Our models allow welfare to vary directly with student type, and show that the ranking of group level gains depends on the taste specification. The magnitudes are not perfectly comparable: Part II computes welfare from fitted mean utilities, while we use utilities recovered by share inversion, including estimated unobserved product quality.

14. Normal vs Finite Type Heterogeneity

Models (a) and (c) estimate near zero standard deviations for some tastes, while the covariance in (d) is numerically singular. Thus, the flexible normal distribution in (d) appears difficult to identify with only aggregate shares. Finite consumer types could capture distinct dietary preferences without imposing a smooth normal distribution. However, our results alone do not show that a finite type model fits better, as we have not estimated one. We would use the simpler diagonal normal model (c) as our main continuous heterogeneity specification and consider finite types as an alternative once there is more individual level information. The average welfare estimates are quite stable across the four models, but the group specific gains are less so.

15. Price Elasticities and Competition

Using the estimates in Question 12, we calculate market level own price and cross price elasticities and then average each entry across campus years in the three product availability periods. With a common price coefficient α, the simulated elasticity is

$$\eta_{jk,mt} = \frac{\hat{\alpha} p_{kmt}}{\hat{s}_{jmt}} \left[\mathbf{1}\{j = k\} \hat{s}_{jmt} - \sum_{h} w_{h,mt} \frac{1}{R} \sum_{r=1}^{R} P_{jrhmt} P_{krhmt} \right].$$

The resulting matrices are 6×6 (2001–2010), 7×7 (2011–2015), and 9×9 (2016–2020). Rows identify the dish whose demand changes and columns the dish whose price changes. Diagonal elasticities are negative and off diagonal elasticities are positive. In 2016–2020, mean own price elasticities range from -0.902 to -0.965 across models (a)–(d), while mean cross price elasticities range from 0.058 to 0.063. Diversion from one gluten free dish toward the other two is about 32.9%–35.0%.

An acquisition of all gluten free dishes by a new firm would internalize their cross product substitution and could soften competition between these dishes. But the original chains would also lose ownership of their gluten free offerings, which changes their pricing incentives. These demand elasticoties by themselves cannot establish the final price or welfare effect without marginal costs and a supply model.

The Part II.9 simple-logit IV estimates imply mean own price elasticities of approximately -0.82, -0.85, and -0.89 across the three periods, compared with -0.83 to -0.88, -0.86 to -0.92, and

−0.90 to −0.97 here. The corresponding mean cross price elasticities under logit IV are about 0.098, 0.077, and 0.057, compared with 0.099–0.106, 0.078–0.083, and 0.058–0.063 under our models. Thus, the RC models imply somewhat stronger price responses but similar overall substitution. Unlike simple logit, they allow substitution to vary with student preferences, although the estimated diversion between gluten free dishes is still close to one third. The nested logit estimates in Part II.10 do not change this conclusion much: the OLS nesting parameter is 0.355, but the IV estimates are close to zero. We therefore would not predict a large difference in merger incentives solely from the choice of demand model. Any actual equilibrium price effect still depends on costs and product ownership.

The full period specific matrices are reported below.

Model (a), 2001-2010: average elasticity matrix.

$j\backslash k$	AN_lm	AN_slm	BN_lm	BN_slm	YN_lm	YN_slm
AN_lm	-0.741	0.086	0.116	0.150	0.080	0.073
AN_slm	0.084	-0.577	0.115	0.159	0.078	0.079
BN_lm	0.087	0.087	-1.030	0.156	0.080	0.076
BN_slm	0.084	0.091	0.117	-0.920	0.079	0.079
YN_lm	0.088	0.087	0.118	0.155	-1.014	0.076
YN_slm	0.084	0.091	0.116	0.160	0.078	-0.760

Model (b), 2001-2010: average elasticity matrix.

$j\backslash k$	AN_lm	AN_slm	BN_lm	BN_slm	YN_lm	YN_slm
AN_lm	-0.717	0.075	0.121	0.137	0.087	0.065
AN_slm	0.074	-0.572	0.113	0.166	0.074	0.082
BN_lm	0.091	0.086	-1.028	0.152	0.080	0.074
BN_slm	0.077	0.094	0.114	-0.915	0.076	0.081
YN_lm	0.096	0.083	0.118	0.149	-1.010	0.073
YN_slm	0.076	0.095	0.113	0.166	0.075	-0.756

Model (c), 2001-2010: average elasticity matrix.

$j\backslash k$	AN_lm	AN_slm	BN_lm	BN_slm	YN_lm	YN_slm
AN_lm	-0.741	0.076	0.133	0.127	0.096	0.066
AN_slm	0.075	-0.588	0.110	0.182	0.073	0.087
BN_lm	0.100	0.084	-1.057	0.149	0.090	0.075
BN_slm	0.071	0.103	0.112	-0.928	0.073	0.089
YN_lm	0.106	0.082	0.133	0.143	-1.039	0.074
YN_slm	0.077	0.102	0.115	0.181	0.076	-0.781

Model (d), 2001-2010: average elasticity matrix.

$j\backslash k$	AN_lm	AN_slm	BN_lm	BN_slm	YN_lm	YN_slm
AN_lm	-0.773	0.078	0.147	0.123	0.104	0.068
AN_slm	0.076	-0.611	0.108	0.204	0.072	0.093
BN_lm	0.111	0.083	-1.089	0.140	0.104	0.077
BN_slm	0.069	0.116	0.105	-0.936	0.069	0.099
YN_lm	0.115	0.081	0.153	0.135	-1.076	0.076
YN_slm	0.079	0.108	0.117	0.200	0.078	-0.814

Model (a), 2011-2015: average elasticity matrix.

$j\backslash k$	AN_lm	AN_slm	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
AN_lm	-0.881	0.067	0.096	0.115	0.064	0.065	0.063
AN_slm	0.072	-0.591	0.094	0.123	0.064	0.063	0.067
BN_lm	0.075	0.069	-1.096	0.121	0.062	0.065	0.066
BN_slm	0.072	0.072	0.096	-0.967	0.062	0.064	0.068
YN_gf	0.075	0.069	0.093	0.115	-0.417	0.063	0.064
YN_lm	0.076	0.068	0.097	0.119	0.063	-1.058	0.065
YN_slm	0.072	0.072	0.096	0.124	0.063	0.063	-1.066

Model (b), 2011-2015: average elasticity matrix.

$j\backslash k$	AN_lm	AN_slm	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
AN_lm	-0.863	0.058	0.098	0.102	0.068	0.071	0.056
AN_slm	0.061	-0.582	0.093	0.132	0.059	0.059	0.073
BN_lm	0.076	0.068	-1.096	0.119	0.063	0.065	0.065
BN_slm	0.064	0.077	0.094	-0.960	0.060	0.060	0.072
YN_gf	0.081	0.064	0.094	0.110	-0.415	0.066	0.061
YN_lm	0.083	0.064	0.098	0.113	0.066	-1.055	0.062
YN_slm	0.063	0.078	0.094	0.130	0.060	0.060	-1.061

Model (c), 2011-2015: average elasticity matrix.

$j\backslash k$	AN_lm	AN_slm	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
AN_lm	-0.891	0.058	0.109	0.097	0.069	0.078	0.057
AN_slm	0.061	-0.601	0.090	0.145	0.063	0.058	0.077
BN_lm	0.085	0.066	-1.127	0.116	0.065	0.073	0.066
BN_slm	0.060	0.085	0.092	-0.977	0.061	0.058	0.078
YN_gf	0.082	0.068	0.097	0.112	-0.428	0.067	0.064
YN_lm	0.091	0.063	0.110	0.109	0.067	-1.086	0.062
YN_slm	0.065	0.082	0.094	0.143	0.063	0.061	-1.099

Model (d), 2011-2015: average elasticity matrix

$j\backslash k$	AN_lm	AN_slm	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
AN_lm	-0.929	0.060	0.122	0.093	0.070	0.086	0.060
AN_slm	0.063	-0.626	0.088	0.161	0.068	0.056	0.081
BN_lm	0.096	0.064	-1.163	0.109	0.065	0.087	0.067
BN_slm	0.058	0.094	0.086	-0.996	0.065	0.055	0.086
YN_gf	0.083	0.074	0.097	0.119	-0.446	0.067	0.067
YN_lm	0.101	0.062	0.129	0.103	0.066	-1.126	0.064
YN_slm	0.068	0.087	0.097	0.156	0.066	0.062	-1.146

Model (a), 2016-2020: average elasticity matrix.

$j\backslash k$	AN_gf	AN_lm	AN_slm	BN_gf	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
AN_gf	-0.552	0.052	0.055	0.051	0.065	0.088	0.057	0.045	0.051
AN_lm	0.058	-1.024	0.051	0.054	0.066	0.085	0.057	0.046	0.048
AN_slm	0.063	0.053	-0.965	0.051	0.065	0.089	0.057	0.045	0.051
BN_gf	0.059	0.056	0.050	-0.420	0.064	0.082	0.058	0.045	0.047
BN_lm	0.060	0.055	0.053	0.051	-1.465	0.090	0.056	0.047	0.050
BN_slm	0.062	0.053	0.055	0.050	0.068	-1.109	0.056	0.046	0.052
YN_gf	0.061	0.055	0.052	0.053	0.064	0.084	-0.570	0.045	0.048
YN_lm	0.060	0.056	0.052	0.052	0.067	0.088	0.056	-1.045	0.050
YN_slm	0.062	0.053	0.055	0.050	0.067	0.091	0.056	0.046	-1.009

Model (b), 2016-2020: average elasticity matrix.

$j\backslash k$	AN_gf	AN_lm	AN_slm	BN_gf	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
AN_gf	-0.546	0.046	0.060	0.044	0.066	0.096	0.054	0.043	0.055
AN_lm	0.051	-1.016	0.044	0.063	0.065	0.076	0.061	0.048	0.043
AN_slm	0.070	0.046	-0.958	0.044	0.066	0.096	0.054	0.043	0.055
BN_gf	0.049	0.066	0.042	-0.408	0.064	0.072	0.061	0.048	0.041
BN_lm	0.061	0.054	0.054	0.051	-1.465	0.089	0.056	0.046	0.050
BN_slm	0.068	0.048	0.059	0.044	0.067	-1.104	0.054	0.044	0.055
YN_gf	0.058	0.058	0.050	0.056	0.065	0.081	-0.568	0.046	0.047
YN_lm	0.057	0.058	0.049	0.056	0.066	0.083	0.058	-1.044	0.048
YN_slm	0.068	0.047	0.060	0.044	0.067	0.096	0.054	0.044	-1.005

Model (c), 2016-2020: average elasticity matrix.

$j\backslash k$	AN_gf	AN_lm	AN_slm	BN_gf	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
AN_gf	-0.565	0.047	0.062	0.050	0.063	0.100	0.058	0.043	0.056
AN_lm	0.052	-1.045	0.044	0.061	0.075	0.071	0.063	0.056	0.044
AN_slm	0.072	0.046	-0.992	0.049	0.064	0.105	0.057	0.043	0.058
BN_gf	0.057	0.063	0.048	-0.431	0.067	0.076	0.062	0.049	0.045
BN_lm	0.059	0.063	0.052	0.054	-1.511	0.088	0.059	0.053	0.051

$j\backslash k$	AN_gf	AN_lm	AN_slm	BN_gf	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
BN_slm	0.071	0.045	0.064	0.047	0.066	-1.131	0.054	0.043	0.060
YN_gf	0.062	0.060	0.052	0.057	0.067	0.082	-0.588	0.048	0.048
YN_lm	0.057	0.067	0.048	0.056	0.076	0.081	0.060	-1.075	0.049
YN_slm	0.069	0.049	0.062	0.048	0.067	0.104	0.056	0.045	-1.041

Model (d), 2016-2020: average elasticity matrix

$j\backslash k$	AN_gf	AN_lm	AN_slm	BN_gf	BN_lm	BN_slm	YN_gf	YN_lm	YN_slm
AN_gf	-0.592	0.049	0.064	0.057	0.063	0.106	0.061	0.043	0.058
AN_lm	0.054	-1.084	0.045	0.055	0.091	0.068	0.065	0.066	0.046
AN_slm	0.075	0.047	-1.036	0.056	0.063	0.115	0.059	0.043	0.061
BN_gf	0.065	0.057	0.054	-0.443	0.060	0.087	0.066	0.043	0.049
BN_lm	0.059	0.077	0.051	0.048	-1.561	0.082	0.060	0.068	0.054
BN_slm	0.075	0.043	0.070	0.054	0.062	-1.163	0.056	0.041	0.065
YN_gf	0.065	0.062	0.055	0.061	0.069	0.085	-0.614	0.049	0.051
YN_lm	0.057	0.080	0.048	0.050	0.096	0.077	0.061	-1.111	0.052
YN_slm	0.071	0.051	0.065	0.053	0.070	0.114	0.058	0.048	-1.085

16. Estimation with Micro Moments

Following Petrin (2002) and Berry, Levinsohn, and Pakes (2004), we re-estimates models (a)–(d) using 10,031 survey responses from 110 campus year markets. All respondents purchased a noodle dish, so we condition observed choice moments on selecting an inside good. First, we add 11 moments consisting of athlete and graduate indicators, their interaction, and each indicator interacted with the four characteristics of the purchased dish. Next, we add 18 moments based on hypothetical second choices: 16 products of first dish and second dish characteristics, one same chain indicator, and one indicator for two gluten free dishes.

For second choices, the first dish is unavailable. The probability of choosing dish $k \neq j$, conditional on having chosen j and student type h, is

$$\Pr(k \mid j, h, mt) = \frac{E_\nu \left[P_{j|h,\nu,mt} \dfrac{e^{V_{k,h,\nu,mt}}}{\sum_{\ell \in \mathcal{J}_{mt} \setminus \{j\}} e^{V_{\ell,h,\nu,mt}}} \right]}{E_\nu[P_{j|h,\nu,mt}]}.$$

The hypothetical choice is restricted to other noodle dishes, excluding the original dish and the outside option. We combine the survey moments with the aggregate IV moments in two step GMM. The two microdata additions are estimated sequentially to show their separate effects. The same aggregate instruments and 512 simulation draws are used.

Table III.5. Re-estimated price coefficients and taste standard deviations

Model	Added microdata	Price	SD spicy	SD wheat	SD veg/meat	SD calories
(a)	Type and choice	-0.850	0.003	0.265	0.588	0.002
(a)	Second choice	-0.860	0.236	0.307	0.704	0.227
(b)	Type and choice	-0.851	–	–	–	–
(b)	Second choice	-0.842	–	–	–	–
(c)	Type and choice	-0.878	0.007	0.067	0.765	0.481
(c)	Second choice	-0.877	0.406	0.160	0.663	0.202
(d)	Type and choice	-0.937	0.606	0.345	0.586	0.498
(d)	Second choice	-0.917	0.445	0.245	0.662	0.604

Table III.6. Athlete and graduate coefficients, with both sets of micro moments

Model	Student type	Spicy	Wheat	Veg/meat	Calories
(b)	Athlete	-0.352	0.798	-0.669	0.698
(b)	Graduate	0.216	-0.127	0.702	-0.559
(c)	Athlete	-0.295	0.880	-0.613	0.720
(c)	Graduate	0.232	-0.093	0.744	-0.473
(d)	Athlete	-0.295	0.906	-0.681	0.707
(d)	Graduate	0.225	-0.115	0.752	-0.422

The demographic choice moments change the athlete and graduate coefficients noticeably compared with aggregate only estimation. The second choice moments also affect taste dispersion: in model (c), the spicy standard deviation changes from about 0.007 to 0.406 when the second choices are included. For model (d), the final correlations between wheat and veg/meat, wheat and calories, and veg/meat and calories are 0.9998, 0.4077, and 0.4236, respectively. Although the additional data distinguish calorie tastes somewhat better, this covariance remains singular to numerical precision.

Table III.7. Welfare and competition by information set

Model	Data	Welfare	Athlete only	Graduate only	Own elasticity	Cross elasticity	GF diversion
(a)	Aggregate	0.480	0.480	0.480	-0.906	0.058	34.1%
(a)	Type-choice	0.480	0.480	0.480	-0.906	0.058	34.1%
(a)	Type+second	0.477	0.477	0.477	-0.914	0.059	34.2%
(b)	Aggregate	0.482	0.607	0.389	-0.902	0.058	32.9%
(b)	Type-choice	0.481	0.377	0.569	-0.905	0.058	34.3%
(b)	Type+second	0.486	0.387	0.561	-0.896	0.057	34.3%
(c)	Aggregate	0.467	0.480	0.412	-0.931	0.060	33.8%
(c)	Type-choice	0.468	0.367	0.553	-0.930	0.060	33.9%
(c)	Type+second	0.471	0.375	0.549	-0.927	0.060	34.4%
(d)	Aggregate	0.453	0.434	0.430	-0.965	0.063	35.0%

Model	Data	Welfare	Athlete only	Graduate only	Own elasticity	Cross elasticity	GF diversion
(d)	Type-choice	0.445	0.350	0.524	-0.985	0.064	35.4%
(d)	Type+second	0.453	0.356	0.534	-0.967	0.063	34.9%

Type choice adds observed choice moments by student type, and Type+second also adds hypothetical choices. Welfare is the 2011–2020 mean compensating variation per lunch choice in price units. Elasticities and gluten-free diversion refer to 2016–2020. These are fixed price counterfactuals.

Average welfare and elasticity estimates change relatively little after adding microdata, but the group specific results change more, which is good as expected. For example, in model (b) the athlete only welfare estimate falls from 0.607 to 0.387, while the graduate only estimate rises from 0.389 to 0.561. Hence the matched demographic data is particularly important for measuring who benefits from new products. The second choice responses further discipline unobserved taste dispersion and correlations, while the estimated substitution between gluten free dishes remains around one third across specifications.